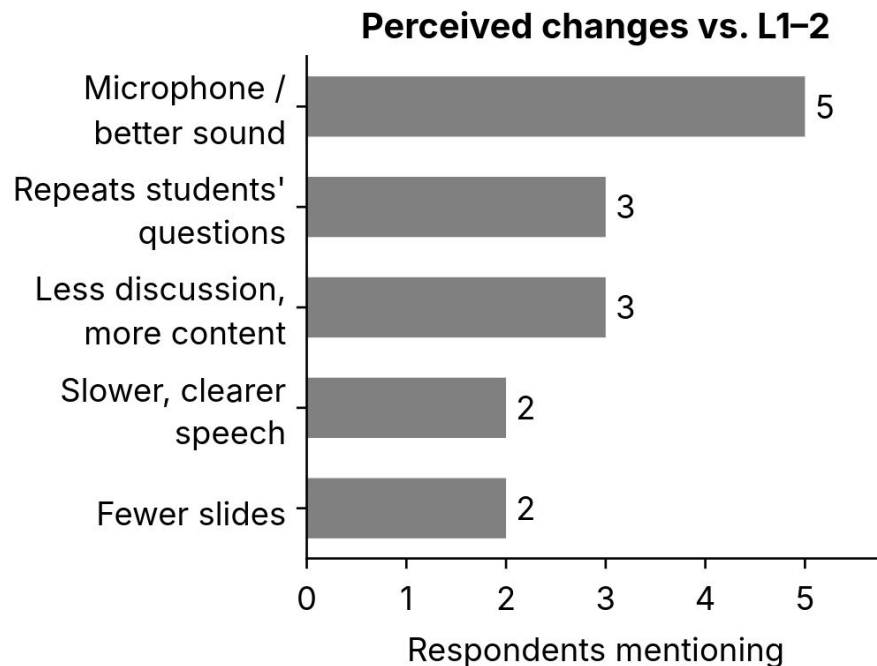
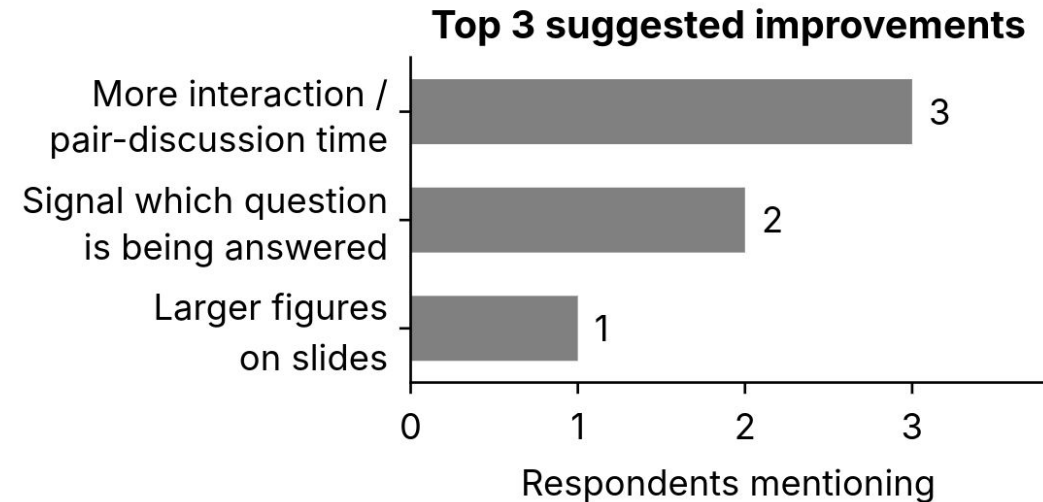
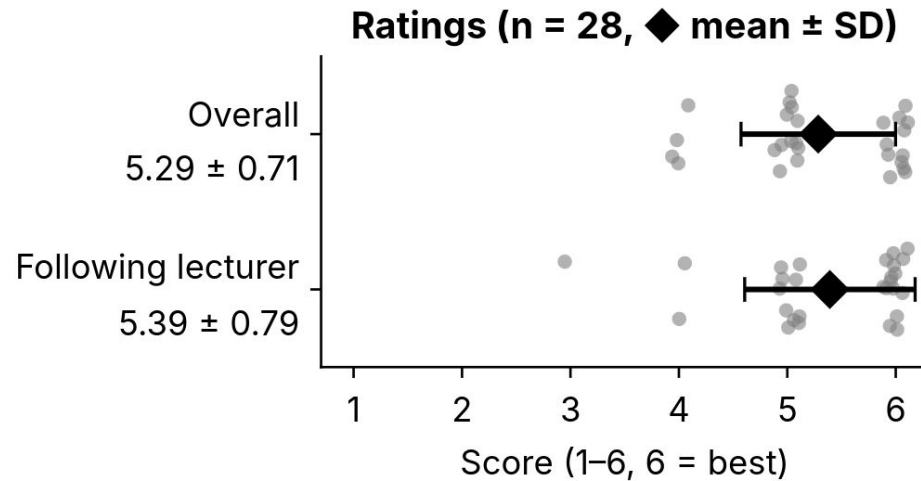


Feedback and questions about lecture 3



1. Why are there no oligonucleotides in the top 20?
2. Why do small molecules and proteins dominate?
3. How do we choose y modality for a target?
4. What does "top selling" measure? Would the balance between SM, LM, and RM look very different if we rank the drugs differently?
5. Why oncology drugs are dominant?

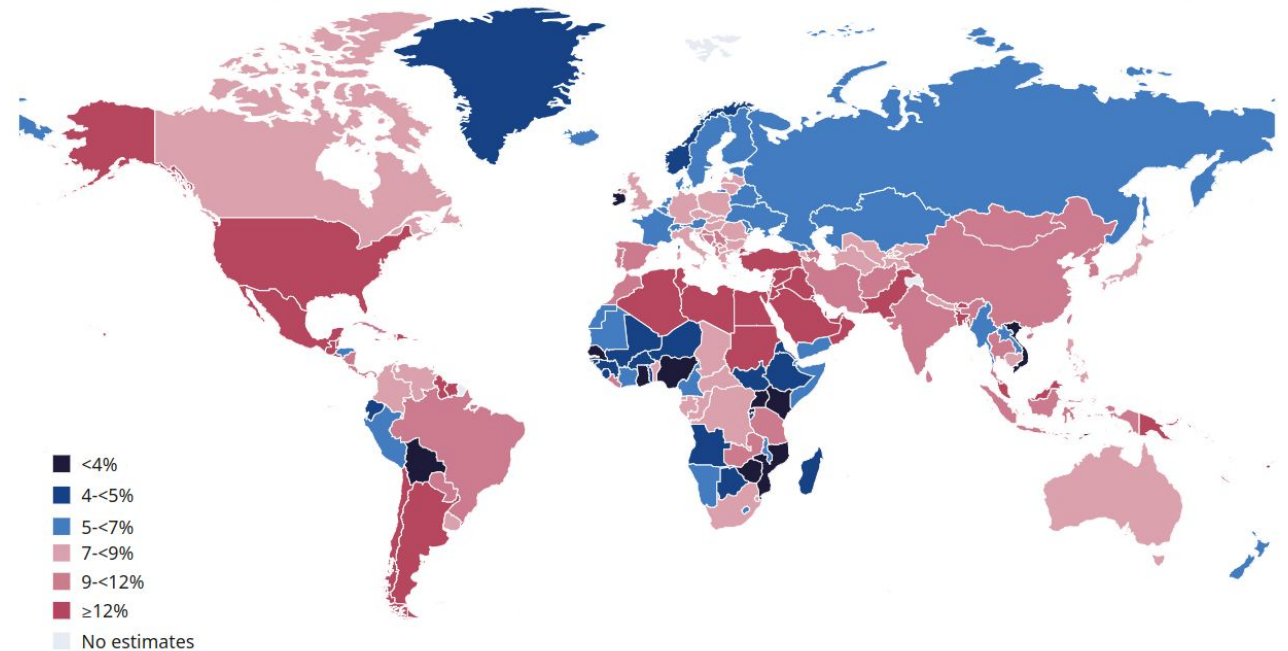
AMIDD 2026 Lecture 4: Key questions in drug discovery

We divide the classroom into seven personas:

1. Basic researchers
2. Research funding agency
3. Patients
4. Medical experts on T2D
5. Drug discovery company
6. Insurance company
7. The regulatory agency

Questions: (1) What are your main interests and concerns? (2) With which groups do you wish to collaborate and why? Rank the partners by priority. (3) What are the best-case and worst-case scenarios for you?

Map 3.2 Estimated age-standardised country prevalence of diabetes in adults (20–79 years), 2024.



Dr. Jitao David Zhang

¹ Computational Sciences Center of Excellence (CS CoE), Roche Innovation Center Basel, F. Hoffmann-La Roche;

² Department of Mathematics and Computer Sciences, University of Basel

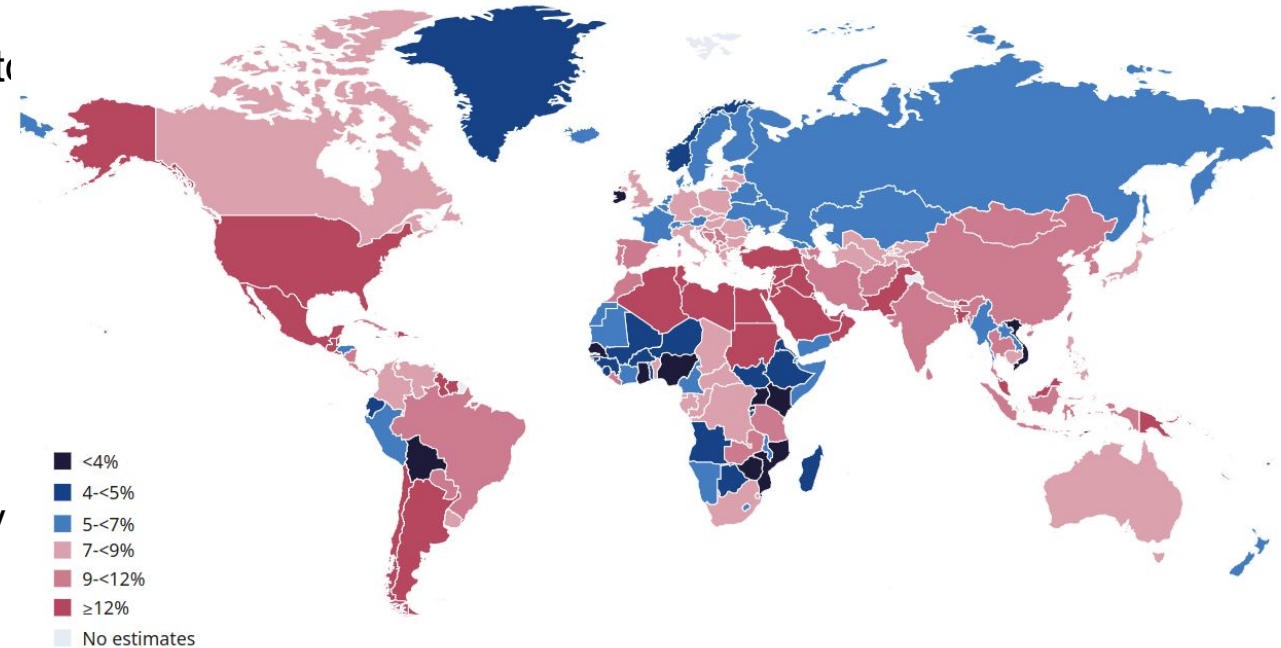
The social context of drug discovery: a role-playing game

We consider a case study of developing new drugs to treat complications caused by type 2 diabetes, affecting ~10% of adults (20–79 y) worldwide (Switzerland: 4-6%, of whom ~28% are not diagnosed), with the possibility to expand it to the obesity and other indications. We divide the classroom into the following personas:

1. Basic researchers (biology, medicine, AI, etc.): doing whatever research you want to do and getting funded
2. Research funding agency: giving fundings to any research problem, T2D is only one of them.
3. Patients: suffering, unfortunately, from T2D and its complications
4. Medical doctors: T2D experts who see patients regularly
5. Drug discovery company: determined to enter the T2D market
6. Insurance company: payer, negotiating the price
7. The regulatory agency: drug approval

Questions: (1) What are your main interests and concerns? (2) With which groups do you wish to collaborate and why? Rank the partners by priority. (3) What are the best-case and worst-case scenarios for you?

Map 3.2 Estimated age-standardised country prevalence of diabetes in adults (20–79 years), 2024.



[IDF Diabetes Atlas 2025](https://diabetesatlas.org/data-by-location/country/switzerland/) (11th version), The International Diabetes Federation (IDF). Swiss data is acquired from <https://diabetesatlas.org/data-by-location/country/switzerland/>

Main interests and concerns

	Interests	Concerns
Basic researcher	Etiology, Novelty, Publication	Funding, Meaningful, Competition
Funding agency	Wellbeing of as many people as possible, Funding research with broad applications,	Reduced total funding, wasted funding.
Patients	Health, Little side effects, insurance coverage	Lack of efficacy, adverse effects, expensive
Medical doctors	Cure, a working health system, work-life balance	Adverse effects, accessibility, long-term unpredictable effects
Pharma company	Working safe drug, IP, high revenue	Scooped, IP infringement, research not translated into drug
Insurance company	Low price, scrutiny over prescription, active market access including generic versions	Over-prescription, high price, people developing chronic disease
Regulatory Agency	Avoid scandals (including corruption), minimizing risk, guideline flexibility	Fake results, protests for drugs not approved, international collaboration

Partners by priority (inputs by the groups)

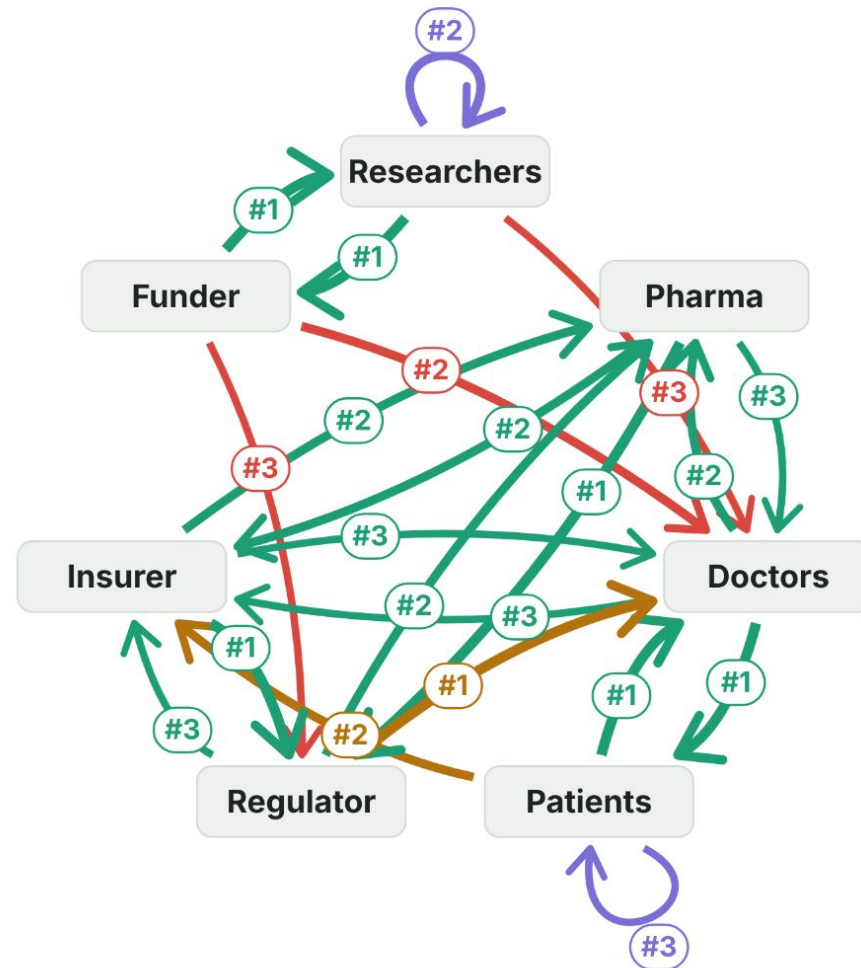


	Priority #1	#2	#3	#4	#5	#6	#7
Basic researcher (A)	F	C	M	P			
Funding agency (F)	C	I	R	A			
Patients (P)	M	I	P	A	R	C	F
Medical doctors (M)	P	C	R	I	F	A	
Pharma company (C)	A	R	M	F	P	I	
Insurance company (I)	R	C	M	P	F	A	
Regulatory Agency (R)	M	P	C	F	A	I	

Partners by priority (inputs by the groups and David)



	Priority #1	#2	#3	#4	#5	#6	#7
Basic researcher (A)	F	A	M	C	P		
Funding agency (F)	A	M	R	I	C		
Patients (P)	M	I	P	C	R	A	
Medical doctors (M)	P	C	I	R	M	F	A
Pharma company (C)	R	I	M	P	C	A	F
Insurance company (I)	R	C	M	P	I		
Regulatory Agency (R)	M	C	I	P	A	R	



- Returned (#1-3)
- Lukewarm (#4-5)
- Unrequited (#6+ or unranked)
- Own peers

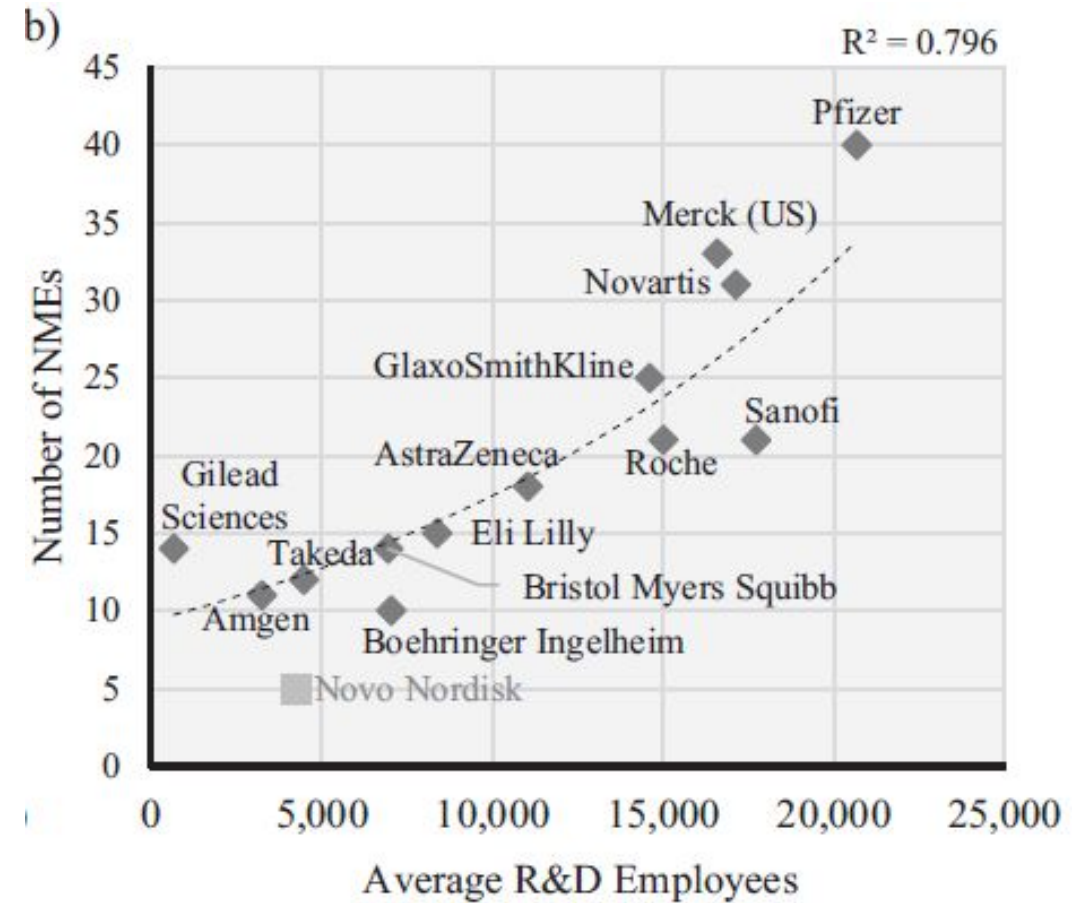
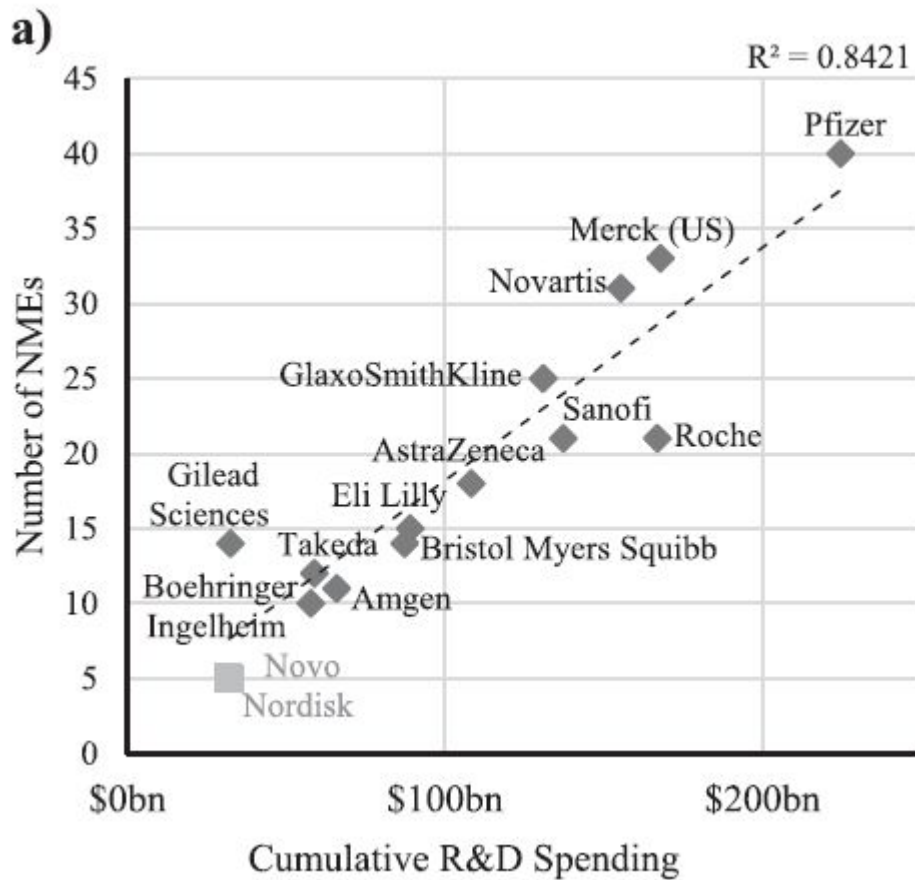
Groups and David: 14 of 19 top-3 picks are returned within the partner's top 3.

Interactive version: <https://claude.ai/artifact/XWrt2TtKb5X412yeJJWnaY>

Best and the worst scenarios

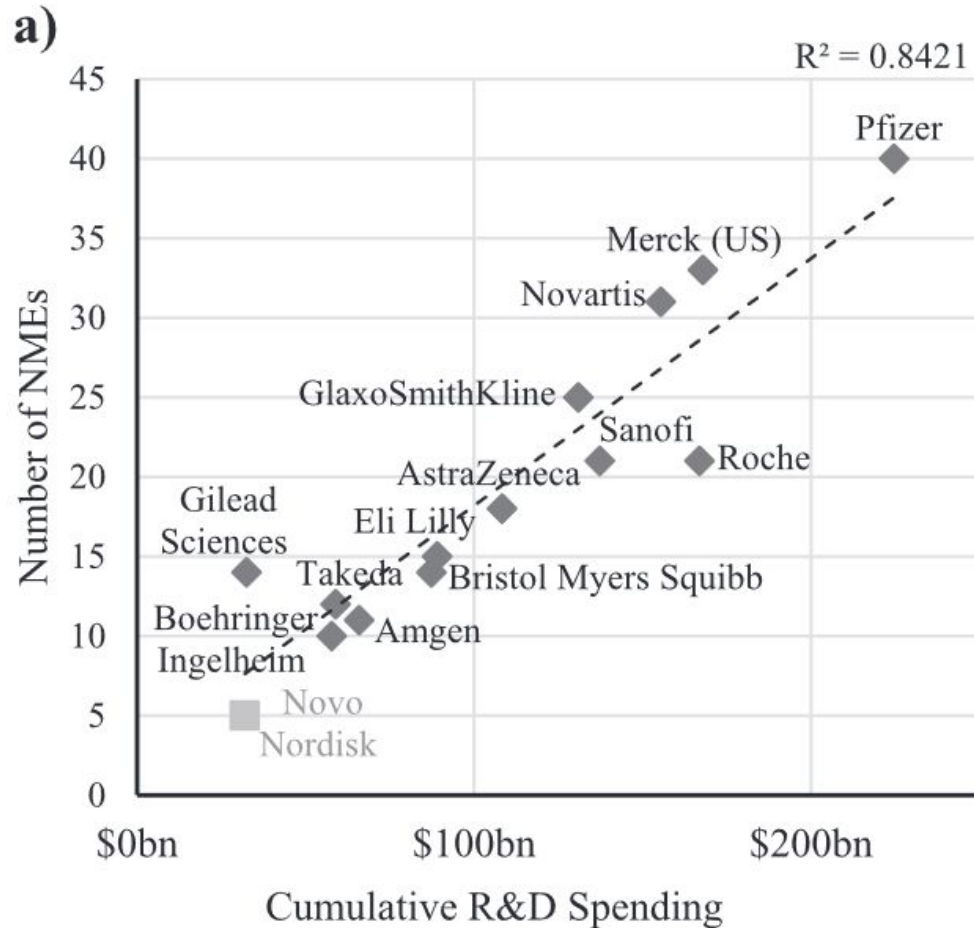
	Ideal scenario	Worst scenario
Basic researchers		
Funding agency		
Patients		
Medical doctors		
Pharma company		
Insurance company		
Regulatory Agency		

Drug discovery and development require huge investment and large interdisciplinary teams

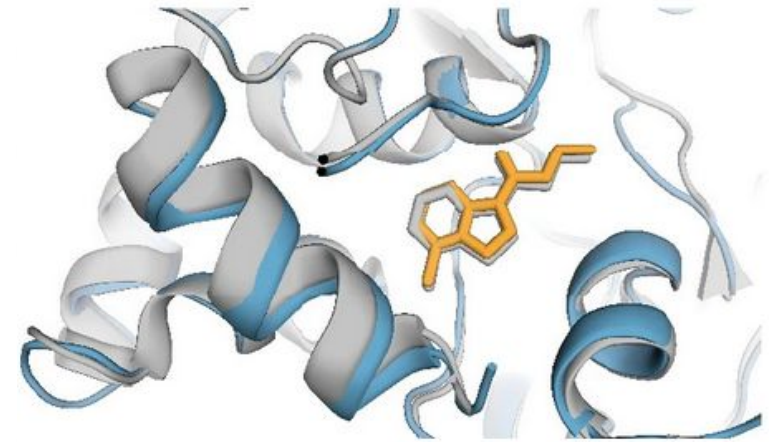
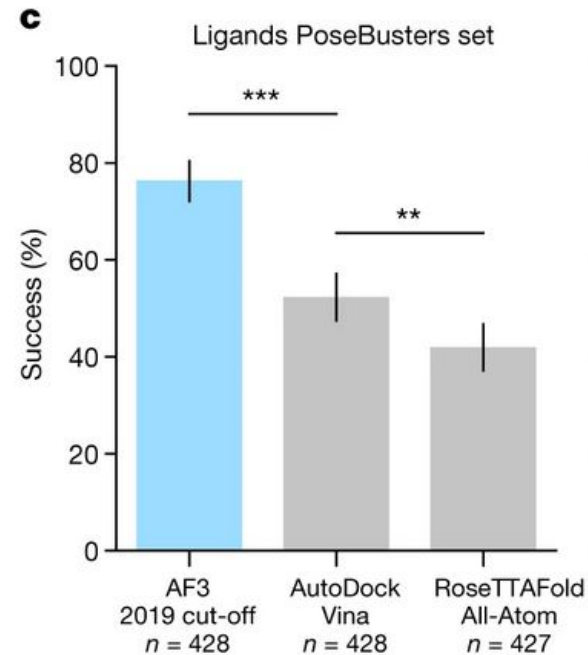


Schuhmacher, Alexander, Lucas Wilisch, Michael Kuss, Andreas Kandelbauer, Markus Hinder, and Oliver Gassmann. "R&D Efficiency of Leading Pharmaceutical Companies – A 20-Year Analysis." *Drug Discovery Today* 26, no. 8 (August 1, 2021): 1784–89. <https://doi.org/10.1016/j.drudis.2021.05.005>.

Quest of the course: to make drug discovery efficient and sustainable with mathematics and informatics



R&D efficiency of leading pharma companies, 1999-2018 (Schumacher *et al.*, 2021)



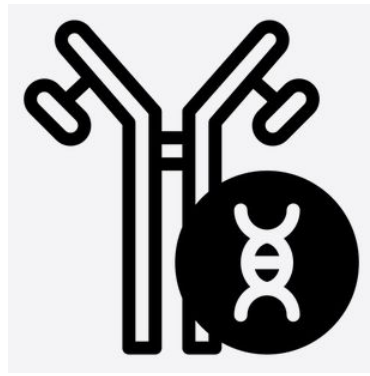
Accurate structure prediction of biomolecular interactions with *AlphaFold3* (Abramson *et al.*, 2024). The *PoseBuster* set: 428 protein-ligand released to PDB after 2021. Success: pocket-aligned ligand Root Mean Square Deviation (RMSD) of atomic positions $\leq 2\text{\AA}$. Right: AF3 prediction for which docking tools *Vina* and *Gold* were less accurate (Human Notum bound to inhibitor ARUK3004556)

Five key questions in drug discovery



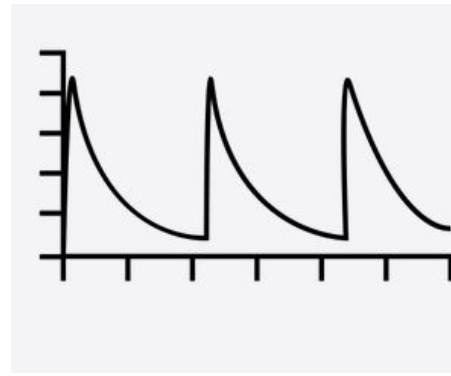
Medical Need

What is the unmet medical need to be addressed?



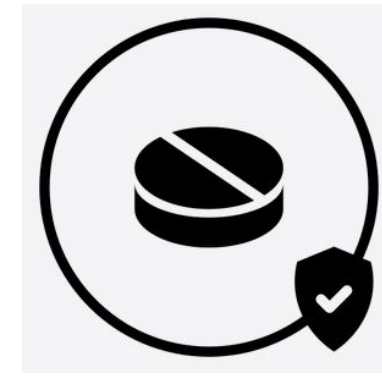
Target & modality

What is the target?
What is the modality?



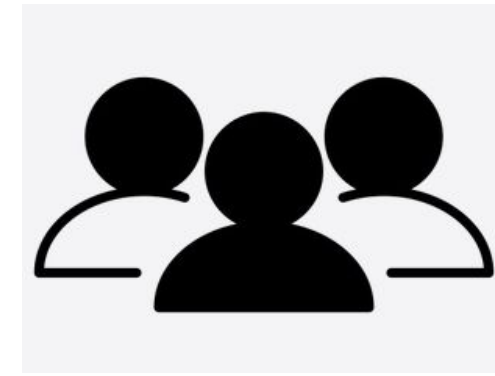
PK/PD

How much drug reach which body part? What does body do to the drug (PK)? What does the drug do to the body (PD)?



Benefit/risk

What is the toxicity of the drug? Is it justifiable given the benefits?

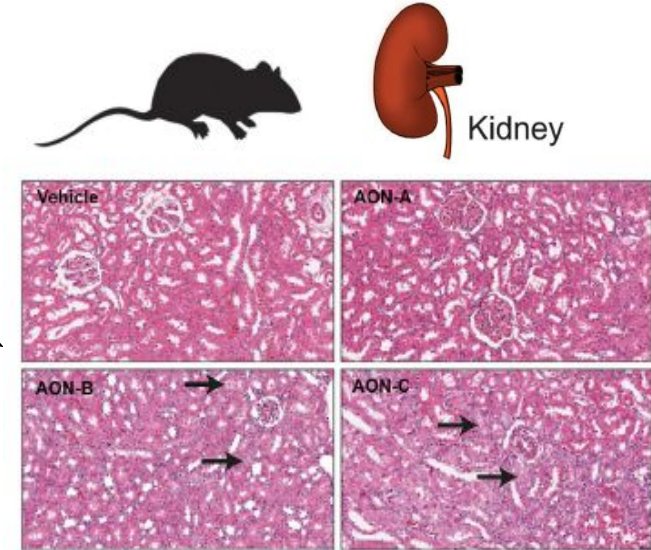
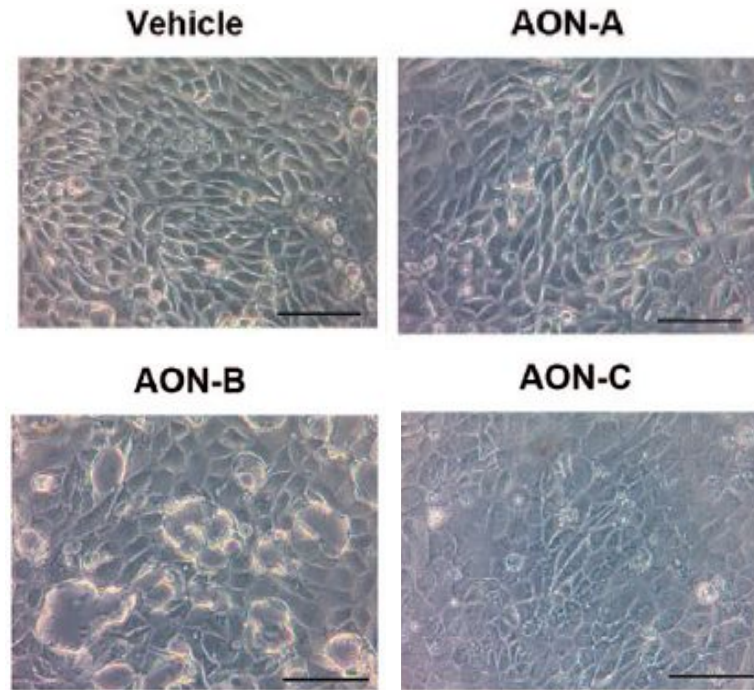
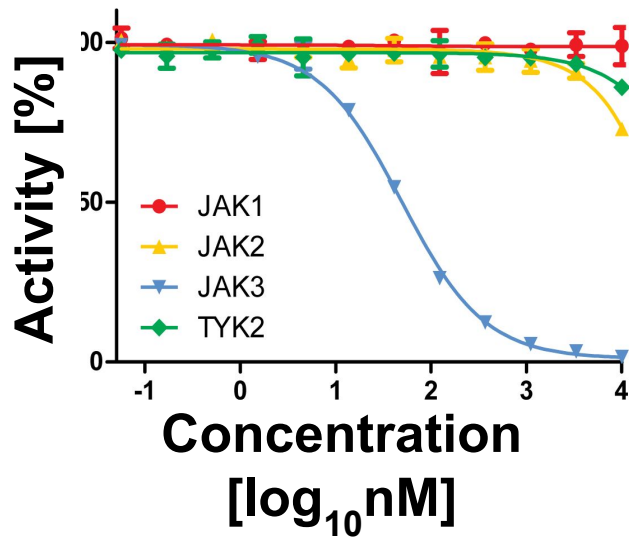


Patient stratification

Who are responsive to the drug? Who are susceptible to adverse events?

Classical workflow of drug discovery from models' perspective

Animal experiments (*in vivo*)



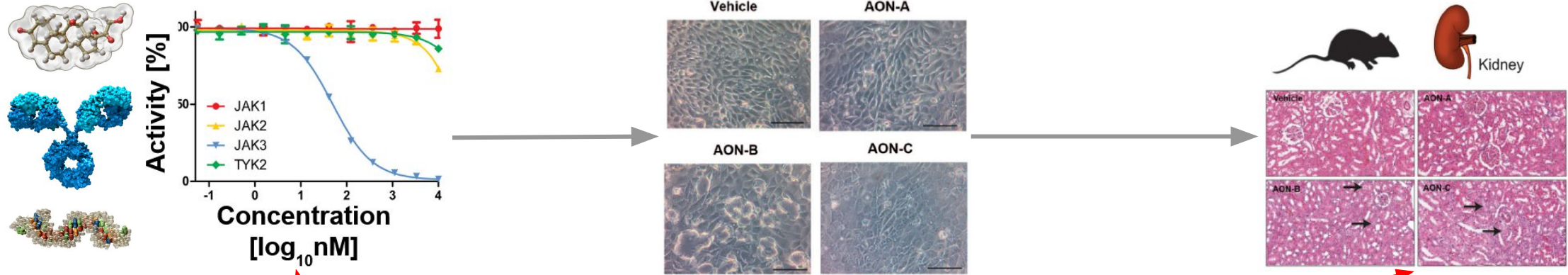
**Biochemical or
biophysical
assays**

**Cellular assays
(*in vitro*)**

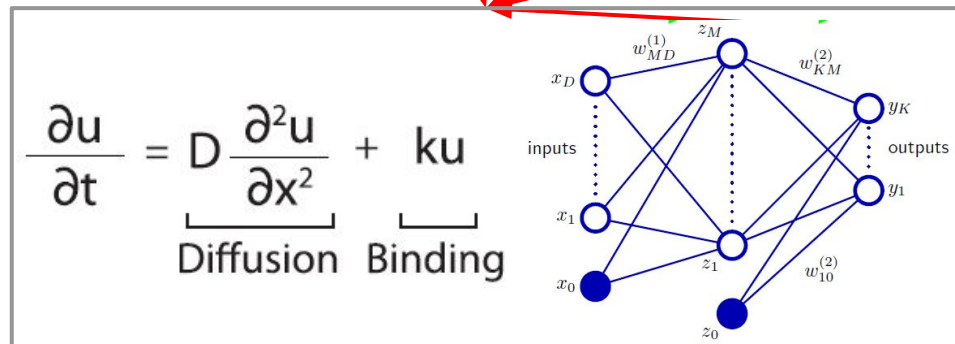


**Clinical
trials**

Mathematical and computational models integrate data across scales



High-throughput technologies (omics, microscopy, etc.)



Mechanistic, causal, and statistical models



Offline activities of Lecture 4

1. Read [the Popular Information of Nobel Prize 2026 in Chemistry](#). What was the most interesting learning for you?
2. Watch the [Nobel Prize Lecture by Katalin Karikó, Nobel Prize Laureate in Physiology or Medicine 2023](#) (42 min), and answer questions.